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Array substrate, method for preparing array substrate, display panel, and display device

Abstract

Provided an array substrate, as well as a method for preparing an array substrate, a display panel, and a display device, includes: substrate; driver circuits located on side of the substrate; signal lead-out structure located on side of the driver circuits facing away from the substrate and including insulating boss and signal lead-out wires one of which electrically connected to an output electrode of a driver circuit, and at least part of the signal lead-out wire extends along sidewall of the insulating boss; a color-resist layer located on side of the driver circuits facing away from the substrate and surrounding sidewall of the signal lead-out structure, and thickness of the color-resist layer less than or equal to extension length of the signal lead-out wire; and a first electrode located on side of the signal lead-out structure facing away from the substrate and electrically connected to the signal lead-out wire.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
(1) This is a national stage application filed under 37 U.S.C. 371 based on International Patent Application No. PCT/CN2021/139027, filed Dec. 17, 2021, which claims priority to Chinese Patent Application No. 202110739102.5 filed Jun. 30, 2021, the disclosures of which are incorporated herein by reference in their entireties.
TECHNICAL FIELD
(2) The present disclosure relates to the field of display technologies, for example, an array substrate, a method for preparing an array substrate, a display panel, and a display device.
BACKGROUND
(3) To meet the demand for comfort and entertainment, there is a trend of large curved screens on the display. For a display device with a large curved screen, due to the design of a color-resist layer having high film thickness, there are problems in that it is difficult to drill a via-hole, that color-resist residues in the via-hole, that wires are prone to be broken during metal climbing, and etc., which affect the process yield and further affect the display effect.
SUMMARY
(4) The present disclosure provides an array substrate, a method for preparing an array substrate, a display panel, and a display device, thus achieving the amelioration of problems in that it is difficult to drill a via-hole, that color-resist residues in the via-hole, that wires are prone to be broken during metal climbing.
(5) An array substrate is provided, including: a substrate; driver circuits located on a side of the substrate, where a driver circuit of the driver circuits includes an output electrode; a signal lead-out structure located on a side of the driver circuits facing away from the substrate, where the signal lead-out structure includes an insulating boss and signal lead-out wires, a signal lead-out wire of the signal lead-out wires is electrically connected to the output electrode, and at least part of the signal lead-out wire extends along a sidewall of the insulating boss; a color-resist layer located on the side of the driver circuits facing away from the substrate, where the color-resist layer surrounds a sidewall of the signal lead-out structure, and where in the first direction, the thickness of the color-resist layer is less than or equal to the extension length of the signal lead-out wire, and the first direction is perpendicular to a plane where the substrate is located; and a first electrode located on a side of the signal lead-out structure facing away from the substrate, where the first electrode is electrically connected to the signal lead-out wire.
(6) A method for preparing an array substrate is further provided, which is used for preparing the above array substrate, and the method includes the following.
(7) A substrate is provided.
(8) Driver circuits are prepared on a side of the substrate, and a driver circuit of the driver circuits includes an output electrode.
(9) A signal lead-out structure is prepared on a side of the driver circuits facing away from the substrate. The signal lead-out structure includes an insulating boss and signal lead-out wires, a signal lead-out wire of the signal lead-out wires is electrically connected to the output electrode, and at least part of the signal lead-out wire extends along a sidewall of the insulating boss.
(10) A color-resist layer is prepared on the side of the driver circuits facing away from the substrate. The color-resist layer surrounds a sidewall of the signal lead-out structure. In the first

direction, the thickness of the color-resist layer is less than or equal to the extension length of the signal lead-out wire, and the first direction is perpendicular to a plane where the substrate is located.
(11) A first electrode is prepared on a side of the signal lead-out structure facing away from the substrate, and the first electrode is electrically connected to the signal lead-out wire.
(12) A display panel is further provided, and the display panel includes the array substrate described above.
(13) A display device is further provided, and the display device includes the display panel described above.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a structural diagram of a display panel;
(2) FIG. **2** is a structural diagram of an array substrate according to an embodiment of the present disclosure;
(3) FIG. **3** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(4) FIG. **4** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(5) FIG. **5** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(6) FIG. **6** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(7) FIG. **7** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(8) FIG. **8** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(9) FIG. **9** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(10) FIG. **10** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(11) FIG. **11** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(12) FIG. **12** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(13) FIG. **13** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(14) FIG. **14** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(15) FIG. **15** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(16) FIG. **16** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(17) FIG. **17** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(18) FIG. **18** is a section view of the array substrate shown in FIG. **14** along a section line A-A′;
(19) FIG. **19** is a section view of the array substrate shown in FIG. **14** along a section line B-B′;
(20) FIG. **20** is another structural diagram of an array substrate according to an embodiment of the

present disclosure;
(21) FIG. **21** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(22) FIG. **22** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(23) FIG. **23** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(24) FIG. **24** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(25) FIG. **25** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(26) FIG. **26** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(27) FIG. **27** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(28) FIG. **28** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(29) FIG. **29** is a structural diagram of an insulating boss according to an embodiment of the present disclosure;
(30) FIG. **30** is another structure diagram of an insulating boss according to an embodiment of the present disclosure;
(31) FIG. **31** is another structure diagram of an insulating boss according to an embodiment of the present disclosure;
(32) FIG. **32** is another structure diagram of an insulating boss according to an embodiment of the present disclosure;
(33) FIG. **33** is another structural diagram of an array substrate according to an embodiment of the present disclosure;
(34) FIG. **34** is a top view of a color-resist layer according to an embodiment of the present disclosure;
(35) FIG. **35** is a flowchart of a method for preparing an array substrate according to an embodiment of the present disclosure;
(36) FIG. **36** is another flowchart of a method for preparing an array substrate according to an embodiment of the present disclosure;
(37) FIG. **37** is another flowchart of a method for preparing an array substrate according to an embodiment of the present disclosure;
(38) FIG. **38** is a structural diagram of a display panel according to an embodiment of the present disclosure; and
(39) FIG. **39** is a structural diagram of a display device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(40) The solutions of the present disclosure are described hereinafter through embodiments in conjunction with the drawings in embodiments of the present disclosure. The embodiments described below are part, not all of the embodiments of the present disclosure.
(41) FIG. **1** is a structural diagram of a display panel. As shown in FIG. **1**, the display panel includes a first substrate **10** and a second substrate **11** which are oppositely disposed, a display dielectric layer **12** disposed between the first substrate **10** and the second substrate **11**, and a color filter structure **13** disposed in the first substrate **10**. The color filter structure **13** may include multiple color-resist blocks having different light-emitting colors. The first substrate **10** may also include driver circuits **14**, and the driver circuits **14** are configured to provide display signals to pixel electrodes **15**, ensuring that the display panel is normally displayed. The driver circuit **14**

includes an active layer, a gate, a source, and a drain, and the pixel electrode **15** is electrically connected to the source or the drain of the driver circuit **14** through a via-hole so that the normal transmission of the display signal is achieved. However, the via-hole has to pass through the color filter structure **13** to achieve the electrical connection between the pixel electrode **15** and the driver circuit **14**. The process is complex, color-resist residues in the via-hole while preparing the via-hole, and the display effect of the display panel is affected. In case that a color-resist layer is relatively thick, it is necessary to form a deep via-hole in the color-resist layer, which can easily cause problems such as that wires are prone to be broken during metal climbing, and the normal display of the display panel is seriously affected.

(42) To solve the above technical problems, embodiments of the present disclosure provide an array substrate. The array substrate include a substrate: driver circuits located on a side of the substrate, where a driver circuit of the driver circuits includes an output electrode: a signal lead-out structure located on a side of the driver circuits facing away from the substrate, where the signal lead-out structure includes an insulating boss and signal lead-out wires, a signal lead-out wire of the signal lead-out wires is electrically connected to the output electrode, and at least part of the signal lead-out wire extends along a sidewall of the insulating boss: a color-resist layer located on the side of the driver circuits facing away from the substrate, where the color-resist layer surrounds a sidewall of the signal lead-out structure, and where in the first direction, the thickness of the color-resist layer is less than or equal to the extension length of the signal lead-out wire, and the first direction is perpendicular to a plane where the substrate is located; and a first electrode located on a side of the signal lead-out structure facing away from the substrate, where the first electrode is electrically connected to the signal lead-out wire. The signal lead-out structure including the insulating boss and the signal lead-out wires is provided, at least part of the signal lead-out wire extends along the sidewall of the insulating boss, and thus the electrical connection between the output electrode of the driver circuit and the first electrode is achieved. That is, the electrical connection between the output electrode and the first electrode is achieved without through a via-hole in the color-resist layer. Therefore, according to the technical solution of the embodiments of the present disclosure, via-holes are not needed to be drilled in the color-resist layer, which makes the preparing process simple, and makes the display not be affected by in-via residues of color-resist, and also makes the problem of wire breakage during metal climbing not be caused by the excessive depth of the via-hole, ensuring good stability of the connection between the output electrode and the first electrode. The thickness of the color-resist layer is less than or equal to the extension length of the signal lead-out wire, that is, the color-resist layer does not completely cover the signal lead-out wire, and thus the electrical connection between the signal lead-out wire and the first electrode is not affected, ensuring the normal transmission of a display signal and normal display of the display panel.

(43) Technical solutions in embodiments of the present disclosure are described hereinafter in conjunction with drawings in the embodiments of the present disclosure.

(44) FIG. **2** is a structural diagram of an array substrate according to an embodiment of the present disclosure, and FIG. **3** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIGS. **2** and **3**, the array substrate **100** provided by the embodiment of the present disclosure includes: a substrate **101**; driver circuits **102** located on a side of the substrate **101**, where a driver circuit **102** includes an output electrode **103**; a signal lead-out structure **104** located on a side of the driver circuits **102** facing away from the substrate **101**, where the signal lead-out structure **104** includes an insulating boss **1041** and signal lead-out wires **1042**, a signal lead-out wire **1042** is electrically connected to the output electrode **103**, and at least part of the signal lead-out wire **1042** extends along a sidewall of the insulating boss **1041**; a color-resist layer **105** located on the side of the driver circuits **102** facing away from the substrate **101**, where the color-resist layer **105** surrounds a sidewall of the signal lead-out structure **104**, and where in the first direction (the direction X shown in the figure), the thickness of the color-resist

layer **105** is less than or equal to the extension length of the signal lead-out wire **1042**, and the first direction is perpendicular to a plane where the substrate **101** is located; and a first electrode **106** located on a side of the signal lead-out structure **104** facing away from the substrate **101**, where the first electrode **106** is electrically connected to the signal lead-out wire **1042**.

(45) The driver circuits **102** are disposed on a side of the substrate **101**, and a driver circuit **102** may include a drive transistor for providing a drive signal to a light-emitting unit. The driver circuit **102** includes the output electrode **103**, and the output electrode **103** may be a source or a drain, which can be selected according to the actual design requirements and is not limited in embodiments of the present disclosure. The signal lead-out structure **104** is located on the side of the driver circuits **102** facing away from the substrate **101**, the color-resist layer **105** is located on the side of the driver circuits **102** facing away from the substrate **101**, and the first electrode **106** is located on the side of the signal lead-out structure **104** facing away from the substrate **101**, that is, the color-resist layer **105** is disposed on a side of the array substrate, i.e., the color filter on array (COA) technology. The color-resist layer **105** is disposed between a film where the first electrode **106** is located and a film where the output electrode **103** is located, as shown in FIG. **2**. In this manner, the light leakage can be ameliorated so that the contrast ratio is increased, and the aperture ratio can be increased at the same time. Due to the color-resist layer **105** located between the film where the first electrode **106** is located and the film where the output electrode **103** is located, in order to avoid drilling a hole in the color-resist layer **105**, the signal lead-out structure **104** is added in the embodiments of the present disclosure. The signal lead-out structure **104** includes the insulating boss **1041** and the signal lead-out wires **1042**, and at least part of a signal lead-out wire **1042** extends along the sidewall of the insulating boss **1041**. The electrical connection between the output electrode **103** and the first electrode **106** is achieved through the insulating boss **1041** and the signal lead-out wires **1042**. The color-resist layer **105** is prepared after the signal lead-out structure **104** is prepared, so that there is no need to drill a via-hole in the color-resist layer **105**, which makes the preparing process simple, and makes the display not be affected by in-via residues of color-resist, and also makes the problem of wire breakage during metal climbing not be caused by the excessive depth of the via-hole, ensuring good stability of the connection between the output electrode **103** and the first electrode **106**.

(46) The insulating boss **1041** may be in a cylindrical shape, a tapered shape, or other shape with a polygonal bottom and a curved sidewall. The shape may be selected according to the actual design requirements and is not limited in the embodiment of the present disclosure. The signal lead-out wires **1042** may be made of a metal having electrical conductivity. At least part of the signal lead-out wire **1042** extends along the sidewall of the insulating boss **1041**, facilitating lead-out of the signal lead-out wire **1042** and reducing the risk of wire breakage of the signal lead-out wire **1042**. The color-resist layer **105** may include color-resist blocks of at least three different colors, such as a red color-resist block, a green color-resist block, and a blue color-resist block. The color-resist layer **105** surrounds the sidewall of the signal lead-out structure **104**, which provides certain fixation and protection to the signal lead-out structure **104**. The first electrode **106** may be a pixel electrode or an anode, which is not limited in the embodiment of the present disclosure.

(47) With further reference to FIGS. **2** and **3**, in the first direction (the direction X shown in FIGS. **2** and **3**), the thickness of the color-resist layer **105** is less than or equal to the extension length of the signal lead-out wire **1042**. In FIG. **2**, it is illustrated an example in which the thickness of the color-resist layer **105** is equal to the extension length of the signal lead-out wire **1042** in the first direction, and in FIG. **3**, it is illustrated an example in which the thickness of the color-resist layer **105** is less than the extension length of the signal lead-out wire **1042** in the first direction, so that the color-resist layer **105** does not completely cover the signal lead-out wire **1042**, that is, the color-resist layer **105** does not affect the electrical connection between the signal lead-out wire **1042** and the first electrode **106**, ensuring that the connection between the signal lead-out wire **1042** and the first electrode **106** is stable.

(48) The array substrate provided by the embodiment of the present disclosure includes the signal lead-out structure located on the side of the driver circuits facing away from the substrate, the color-resist layer located on the side of the driver circuits facing away from the substrate, and the first electrode located on the side of the signal lead-out structure facing away from the substrate. The signal lead-out structure includes the insulating boss and the signal lead-out wires, and at least part of the signal lead-out wire extends along the sidewall of the insulating boss. The electrical connection between the output electrode of the driver circuit and the first electrode is achieved by using the signal lead-out structure, there is no need to drill a via-hole in the color-resist layer, which makes the preparing process simple, and makes the display not be affected by in-via residues of color-resist, and also makes the problem of wire breakage during metal climbing not be caused by the excessive depth of the via-hole, ensuring good stability of the connection between the output electrode and the first electrode, the normal transmission of the display signal and normal display of the display panel.
(49) In some embodiments, FIG. **4** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **5** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **6** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **7** is another structural diagram of an array substrate according to an embodiment of the present disclosure, and FIG. **8** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIGS. **4** to **8**, the color-resist layer **105** includes multiple color-resist blocks **1051**, and at least one color-resist block **1051** contacts a sidewall of the signal lead-out structure **104**.
(50) The color-resist layer **105** includes multiple color-resist blocks **1051**, and color-resist blocks of different colors can emit light in different colors.
(51) At least one color-resist block **1051** contacts the sidewall of the signal lead-out structure **104**, that is, as shown in FIG. **4**, it may be that one color-resist block **1051** contacts the sidewall of the signal lead-out structure **104**; and as shown in FIGS. **5** to **8**, it may also be that multiple color-resist blocks **1051** contact the sidewall of the signal lead-out structure **104**. As shown in FIG. **4**, in a case where one color-resist block **1051** contacts the sidewall of the signal lead-out structure **104**, the signal lead-out structure **104** may be located within the coverage area of the color-resist layer **105**. As shown in FIGS. **5** to **8**, in a case where multiple color-resist blocks **1051** contact the sidewall of the signal lead-out structure **104**, the signal lead-out structure **104** may be located in a region in which multiple color-resist blocks **1051** are close to each other. For example, in FIG. **5**, it is illustrated three types of color-resist blocks **1051**, each having a quadrilateral shape, which share a same signal lead-out structure **104** having a hexagonal shape. For example, in FIG. **6**, it is illustrated two types of color-resist blocks **1051**, each having a hexagonal shape, which share a same signal lead-out structure **104** having a hexagonal shape. For example, in FIG. **7**, it is illustrated two types of color-resist blocks **1051**, each having a quadrilateral shape, which share the same signal lead-out structure **104** having an octagonal shape. For example, in FIG. **8**, it is illustrated six types of color-resist blocks **1051**, each having a triangular shape, which share the same signal lead-out structure **104** having a dodecagonal shape. Multiple color-resist blocks **1051** contact different sidewalls of the same signal lead-out structure **104**, so that the number of insulating bosses **1041** in the signal lead-out structure **104** can be reduced, the coverage area of the color-resist layer **105** in the array substrate can be increased, and with this arrangement, the display aperture ratio of the display panel can be increased.
(52) FIG. **9** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIG. **9**, in some embodiments, at least two color-resist blocks **1051** surround the sidewall of the signal lead-out structure **104**.
(53) The driver circuits **102** include multiple driver circuit sets, each driver circuit set includes at least a first driver circuit **1021** and a second driver circuit **1022**, and the number of driver circuits in

each driver circuit set is the same as the number of color-resist blocks **1051** around the sidewall of the signal lead-out structure **104**. The first driver circuit **1021** includes a first output electrode **1031**, and the second driver circuit **1022** includes a second output electrode **1032**.
(54) The signal lead-out wires **1042** include at least a first signal lead-out wire **1043** and a second signal lead-out wire **1044**, and the first signal lead-out wire **1043** is insulated from the second signal lead-out wire **1044**.
(55) The first signal lead-out wire **1043** is electrically connected to the first output electrode **1031**, and the second signal lead-out wire **1044** is electrically connected to the second output electrode **1032**.
(56) As shown in FIG. **9**, in a case where at least two color-resist blocks **1051** surround the sidewall of a signal lead-out structure **104**, output electrodes **103** of driver circuits, which corresponds to at least two color-resist blocks **1051** surrounding the sidewall of the signal lead-out structure **104**, each is electrically connected to a corresponding first electrode **106** through the signal lead-out structure **104**. For example, two color-resist blocks **1051** surrounding the sidewalls of the signal lead-out structure **104** are shown in FIG. **9**. Correspondingly, a driver circuit set includes two driver circuits **102**, namely the first driver circuit **1021** and the second driver circuit **1022** as shown in FIG. **9**, and the signal lead-out wires **1042** includes two signal lead-out wires **1042** insulated from each other, namely the first signal lead-out wire **1043** and the second signal lead-out wire **1044** as shown in FIG. **9**. The first driver circuit **1021** includes the first output electrode **1031** and the second driver circuit **1022** includes the second output electrode **1032**. The first output electrode **1031** is electrically connected to a corresponding first electrode **106** through the first signal lead-out wire **1043**, the second output electrode **1032** is electrically connected to a corresponding first electrode **106** through the second signal lead-out wire **1044**, and thus the electrical connection between a driver circuit and a corresponding first electrode **106** is achieved and it is ensured that the first electrode **106** receives the display signal normally. In this manner, the signal lead-out structure **104** is disposed in a region in which multiple color-resist blocks **1051** are close to each other, and multiple signal lead-out wires **1042**, which are insulated from each other, are arranged to extend along the sidewall of the insulating boss **1041**, which can ensure that multiple driver circuits are connected to corresponding first electrodes **106** through one insulating boss **1041**, and thus, the number of insulating bosses **1041** can be reduced, the coverage area of the color-resist layer **105** in the array substrate can be increased, and the display aperture ratio of the display panel can be increased in this arrangement.
(57) Based on the above-mentioned embodiments, in a case where at least two color-resist blocks **1051** surround the sidewall of the signal lead-out structure **104**, there may be various different arrangements of driver circuit sets or signal lead-out wires **1042**. Two feasible arrangements are described as examples in the following, to illustrate how to achieve the electrical connection between a driver circuit set and a corresponding first electrode **106** in a case where at least two color-resist blocks **1051** surround the sidewall of the signal lead-out structure **104**.
(58) As a feasible embodiment, with further reference to FIG. **9**, in some embodiments, the driver circuit set is arranged to surround a signal lead-out structure **104**.
(59) The position of driver circuits **102** in the driver circuit set may be adjusted. For example, the driver circuit set is arranged to surround the signal lead-out structure **104**, so that different driver circuits **102** in the driver circuit set share a same insulating boss **1041** of the same signal lead-out structure **104**, thereby ensuring that the number of insulating bosses **1041** in the signal lead-out structures **104** can be reduced and the coverage area of the color-resist layer **105** can be increased. As an example, in FIG. **9**, the first driver circuit **1021** and the second driver circuit **1022** in the driver circuit set are arranged to surround the signal lead-out structure **104**, and the first signal lead-out wire **1043** connected to the first output electrode **1031** with the first electrode **106** and the second signal lead-out wire **1044** connected to the second output electrode **1032** with the first electrode **106** are arranged to extend along the sidewall of the same insulating boss **1041**. Thus, on

the premise of ensuring the normal transmission of the drive signal, the coverage area of the color-resist layer **105** is increased, and the display aperture ratio of the display panel can be increased in this arrangement.
(60) Based on the above-described embodiments, how a data line and a scan line are connected to a driver circuit in a case where a driver circuit set is arranged to surround a signal lead-out structure **104** is described in the following.
(61) In some embodiments, FIG. **10** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **11** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **12** is another structural diagram of an array substrate according to an embodiment of the present disclosure, and FIG. **13** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIGS. **10** to **13**, FIGS. **10** to **13** only illustrate the connection as an example in a case where one driver circuit set **102** is arranged to surround the signal lead-out structure **104**, and the same connection in the same array substrate is not repeated herein. Multiple color-resist blocks **1051** are arranged in a matrix, multiple color-resist blocks **1051** include multiple color-resist columns **107**, and a first color-resist gap **1071** exists between two adjacent color-resist columns **107**.
(62) The array substrate **100** further includes multiple data signal line sets **108**, a data signal line set **108** includes at least a first data signal line **1081** and a second data signal line **1082**, the first data signal line **1081** is electrically connected to the first driver circuit **1021**, and the second data signal line **1082** is electrically connected to the second driver circuit **1022**.
(63) A vertical projection of the first data signal line **1081** on the plane where the substrate **101** is located and a vertical projection of the second data signal line **1082** on the plane where the substrate **101** is located overlap with vertical projections of different first color-resist gaps **1071** on the plane where the substrate **101** is located, respectively. The first data signal line **1081** further includes a first data connection line **1083**, and the first driver circuit **1021** is electrically connected to the first data signal line **1081** via the first data connection line **1083**, or the second data signal line **1082** further includes a first data connection line **1083**, and the second driver circuit **1022** is electrically connected to the second data signal line **1082** via the first data connection line **1083**.
(64) Multiple color-resist blocks **1051** are arranged in a matrix, multiple color-resist blocks **1051** include multiple color-resist columns **107**, the first color-resist gap **1071** exists between two adjacent color-resist columns **107**, and the first color-resist gap **1071** avoids optical crosstalk between adjacent color-resist columns **107**. The array substrate **100** further includes multiple data signal line sets **108** for providing driver circuits **102** with data signals. Taking the structure shown in FIGS. **10** to **13** as an example, the data signal line set **108** includes the first data signal line **1081** and the second data signal line **1082**. The vertical projection of the first data signal line **1081** on the plane where the substrate **101** is located and the vertical projection of the second data signal line **1082** on the plane where the substrate **101** is located overlap with the vertical projections of different first color-resist gaps **1071** on the plane where the substrate **101** is located, respectively, that is, the first data signal line **1081** and the second data signal line **1082** overlap with different first color-resist gaps **1071** in the direction perpendicular to the plane where the substrate **101** is located, so that crosstalk between data signal lines can be effectively avoided.
(65) To ensure that the first data signal line **1081** and the second data signal line **1082** are electrically connected to driver circuits **102** which correspond to three adjacent color-resist columns **107**, the first data signal line **1081** or the second data signal line **1082** further includes the first data connection line **1083**. In FIGS. **10** to **12**, that the second data signal line **1082** includes the first data connection line **1083** is taken as an example, and in FIG. **13**, that the first data signal line **1081** includes the first data connection line **1083** is taken as an example. As shown in FIGS. **10** to **13**, the first driver circuit **1021** is electrically connected to the first data signal line **1081** via the first data connection line **1083**, thus achieving that the first driver circuit **1021** being relatively far from

the first data signal line **1081** can receive a data signal outputted by the first data signal line **1081**, achieving that the first driver circuit **1021** outputs the drive signal and achieving the display. Similarly, the second driver circuit **1022** may be electrically connected to the second data signal line **1082** via the first data connection line **1083**, thereby ensuring normal display.
(66) To describe the connection between the driver circuits **102** and the data signal lines, only the driver circuits **102** in the region corresponding to part of color-resist blocks **1051**, rather than all the driver circuits **102** in the region corresponding to all of color-resist blocks **1051** are shown in FIGS. **10** to **14**. The arrangement of remaining driver circuits **102** is similar to that of the driver circuits **102** shown in the FIGS. **10** to **14**, and details are not repeated herein. Only part of driver circuits **102** are shown in subsequent drawings too, and the arrangement of the remaining driver circuits **102** is also similar to that of the driver circuits **102** shown in the subsequent drawings.
(67) In summary, in the above-mentioned embodiments, different data signal lines overlap with different first color-resist gaps **1071** respectively, and how to achieve the connection between a data signal line and a driver circuit **102** is implemented by adding the first data connection line **1083**. Next, the connection between a data signal line and a driver circuit **102** is described in another embodiment.
(68) FIG. **14** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **15** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **16** is another structural diagram of an array substrate according to an embodiment of the present disclosure, and FIG. **17** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIGS. **14** to **17**, in some embodiments, multiple color-resist blocks **1051** are arranged in a matrix, multiple color-resist blocks **1051** include multiple color-resist columns **107**, and a first color-resist gap **1071** exists between two adjacent color-resist columns **107**.
(69) The array substrate **100** further includes multiple data signal line sets **108**, a data signal line set **108** includes at least a first data signal line **1081** and a second data signal line **1082**, the first data signal line **1081** is electrically connected to the first driver circuit **1021**, and the second data signal line **1082** is electrically connected to the second driver circuit **1022**.
(70) A vertical projection of the first data signal line **1081** on the plane where the substrate **101** is located and a vertical projection of the second data signal line **1082** on the plane where the substrate **101** is located overlap with a vertical projection of a same first color-resist gap **1071** on the plane where the substrate **101** is located.
(71) Taking the structure shown in FIGS. **14** to **17** as an example, each data signal line set **108** includes the first data signal line **1081** and the second data signal line **1082**, the first data signal line **1081** is electrically connected to the first driver circuit **1021**, and the second data signal line **1082** is electrically connected to the second driver circuit **1022**, thus achieving transmission of a data signal in the driver circuit **102**, thereby ensuring the display effect. The vertical projection of the first data signal line **1081** on the plane where the substrate **101** is located and the vertical projection of the second data signal line **1082** on the plane where the substrate **101** is located overlap with the vertical projection of the same first color-resist gap **1071** on the plane where the substrate **101** is located, that is, the first data signal line **1081** and the second data signal line **1082** overlap with the same first color-resist gaps **1071** in the direction perpendicular to the plane where the substrate **101** is located, so that the first data signal line **1081** and the second data signal line **1082** can be directly electrically connected to corresponding driver circuits. In this manner, the drive of driver circuits corresponding to adjacent color-resist columns **107** can be achieved without arranging a first data connection line **1083**, thereby ensuring a simple connection between data signal lines and corresponding driver circuits.
(72) Only one kind of connection between driver circuits **102** and data signal line sets **108** is shown in FIGS. **10** to **17** as an example. Other connections may be selected according to actual design requirements and are not limited in the embodiments.

(73) In an example, blocks represent driver circuits in FIGS. **10** to **17** and subsequent drawings, and embodiments of the present disclosure do not limit the shape and structure of the driver circuits. Furthermore, in order to represent the electrical connection between a driver circuit and one of a data signal line, a scan signal line, or the signal lead-out structure, examples also show that the projection of the driver circuit does not overlap with each projection of projections of the data signal line, the scan signal line, and the signal lead-out structure, and the electrical connection is achieved through a connection line. In an actual structure, part of the films in the driver circuits may be disposed in the same layer with the data signal line and the scan signal line. The projection of the driver circuit may overlap with the projection of the data signal line, and the projection of the driver circuit may overlap with the projection of the scan signal line. The connection may be achieved by using a via-hole and there is no connection line, which is not limited in embodiments of the present disclosure.
(74) FIG. **18** is a section view of the array substrate shown in FIG. **14** along a section line A-A′, and FIG. **19** is a section view of the array substrate shown in FIG. **14** along a section line B-B′. As shown in FIG. **14**, FIG. **18**, and FIG. **19**, in some embodiments, the first data signal line **1081** and the second data signal line **1082** are disposed in different layers.
(75) The vertical projection of the first data signal line **1081** on the plane where the substrate **101** is located and the vertical projection of the second data signal line **1082** on the plane where the substrate **101** is located overlap with the vertical projection of the same first color-resist gap **1071** on the plane where the substrate **101** is located. In general, the first data signal line **1081** and the second data signal line **1082** transmit different data signals. To avoid crosstalk between the first data signal line **1081** and the second data signal line **1082** which are adjacent, the first data signal line **1081** and the second data signal line **1082** may be disposed in different layers, thereby ensuring the normal transmission of the data signal.
(76) In some embodiments, with further reference to FIGS. **18** and **19**, the first data signal line **1081** is located on a side of the color-resist layer **105** facing to the substrate **101**, and the second data signal line **1082** is located on a side of the color-resist layer **105** facing away from the substrate **101**.
(77) The second data signal line **1082** further includes a second data connection line **1084**, the second data signal line **1082** is electrically connected to the second driver circuit **1022** via the second data connection line **1084**, and the second data connection line **1084** extends along a sidewall of the insulating boss **1041**.
(78) In general, the first data signal line **1081** and the output electrode **103** of the driver circuit **102** are disposed in the same layer, and namely the first data signal line **1081** is located on the side of the color-resist layer **105** facing to the substrate **101**. Since the first data signal line **1081** and the second data signal line **1082** are disposed in different layers, the second data signal line **1082** may be located on the side of the color-resist layer **105** facing away from the substrate **101**, the second data signal line **1082** is electrically connected to the second driver circuit **1022** via the second data connection line **1084**, and the second data connection line **1084** extends along the sidewall of the insulating boss **1041**. Namely, the arrangement of the second data connection line **1084** may be the same as the arrangement of the signal lead-out wire **1042**. In this manner, the process of drilling via-holes and residues in the via-holes can be avoided when the second data signal line **1082** is electrically connected to a corresponding driver circuit, and the electrical connection between the second data signal line **1082** and the second driver circuit **1022** is achieved.
(79) In summary, the data signal lines are taken as an example to describe the arrangement of the data signal lines in a case where the driver circuit sets are arranged to surround the signal lead-out structure, and the arrangement of the scan signal lines is taken as an example in the following.
(80) FIG. **20** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **21** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **22** is another structural diagram of an array substrate

according to an embodiment of the present disclosure, and FIG. **23** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIGS. **20** to **23**, in some embodiments, multiple color-resist blocks **1051** are arranged in a matrix, multiple color-resist blocks **1051** include multiple color-resist rows **109**, and a second color-resist gap **1091** exists between two adjacent color-resist rows **109**.
(81) The array substrate **100** further includes multiple scan signal line sets **110**, a scan signal line set **110** includes at least a first scan signal line **1101** and a second scan signal line **1102**, the first scan signal line **1101** is electrically connected to the first driver circuit **1021**, and the second scan signal line **1102** is electrically connected to the second driver circuit **1022**.
(82) A vertical projection of the first scan signal line **1101** on the plane where the substrate **101** is located and a vertical projection of the second scan signal line **1102** on the plane where the substrate **101** is located overlap with vertical projections of different second color-resist gaps **1091** on the plane where the substrate **101** is located, respectively; and the first scan signal line **1101** further includes a first scan connection line **1103**, and is electrically connected to the first scan signal line **1101** via the first scan connection line **1103**, or the second scan signal line **1102** further includes a first scan connection line **1103**, and is electrically connected to the second scan signal line **1102** via the first scan connection line **1103**.
(83) Multiple color-resist blocks **1051** are arranged in a matrix, multiple color-resist blocks **1051** include multiple color-resist rows **109**, the second color-resist gap **1091** exists between two adjacent color-resist rows **109**, and optical crosstalk between adjacent color-resist rows **109** is avoided by the second color-resist gap **1091**. Taking the structure shown in FIGS. **20** to **23** as an example, a scan signal line set **110** including the first scan signal line **1101** and the second scan signal line **1102** is described as an example, the first scan signal line **1101** is electrically connected to the first driver circuit **1021**, and the second scan signal line **1102** is electrically connected to the second driver circuit **1022**, thus achieving transmission of a scan signal in the driver circuit **102**, thereby ensuring the display effect. The vertical projection of the first scan signal line **1101** on the plane where the substrate **101** is located and the vertical projection of the second scan signal line **1102** on the plane where the substrate **101** is located overlap with the vertical projections of different second color-resist gaps **1091** on the plane where the substrate **101** is located, respectively, that is, the first scan signal line **1101** and the second scan signal line **1102** overlap with different second color-resist gaps **1091** in the direction perpendicular to the plane where the substrate **101** is located, thus avoiding the crosstalk between the first scan signal line **1101** and the second scan signal line **1102**.
(84) To ensure that each scan signal line performs signal scanning on a transmission of each color-resist row **109**, the first scan signal line **1101** or the second scan signal line **1102** further includes the first scan connection line **1103**. In FIGS. **20**, **22** to **23**, the first scan signal line **1101** including the first scan connection line **1103** is described as an example, and in FIG. **21** the second scan signal line **1102** including the first scan connection line **1103** is described as an example. As shown in FIGS. **20** to **23**, the first driver circuit **1021** is electrically connected to the first scan signal line **1101** via the first scan connection line **1103**, thus achieving that a scan signal of the first scan signal line **1101** can be transmitted to the first driver circuit **1021** and the first driver circuit **1021** can output a corresponding drive signal, thereby ensuring the display. Similarly, the second driver circuit **1022** may be electrically connected to the second scan signal line **1102** via the first scan connection line **1103**, thereby ensuring the display effect.
(85) In summary, in the above-mentioned embodiments, different scan signal lines overlap with different second color-resist gaps, and how to achieve the connection between a scan signal line and a driver circuit is explained by adding the first scan connection line. Another embodiment for the connection between a scan signal line and a driver circuit is described in the following.
(86) FIG. **24** is another structural diagram of an array substrate according to an embodiment of the present disclosure, FIG. **25** is another structural diagram of an array substrate according to an

embodiment of the present disclosure, FIG. **26** is another structural diagram of an array substrate according to an embodiment of the present disclosure, and FIG. **27** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIGS. **24** to **27**, in some embodiments, multiple color-resist blocks **1051** are arranged in a matrix, multiple color-resist blocks **1051** include multiple color-resist rows **109**, and a second color-resist gap **1091** exists between two adjacent color-resist rows **109**.
(87) The array substrate **100** further includes multiple scan signal line sets **110**, each scan signal line set **110** includes at least a first scan signal line **1101** and a second scan signal line **1102**, the first scan signal line **1101** is electrically connected to the first driver circuit **1021**, and the second scan signal line **1102** is electrically connected to the second driver circuit **1022**.
(88) The vertical projection of the first scan signal line **1101** on the plane where the substrate **101** is located and the vertical projection of the second scan signal line **1102** on the plane where the substrate **101** is located overlap with a vertical projection of the same second color-resist gap **1091** on the plane where the substrate **101** is located.
(89) Taking the structure shown in FIGS. **24** to **27** as an example, a scan signal line set **110** includes a first scan signal line **1101** and a second scan signal line **1102**, the first scan signal line **1101** is electrically connected to the first driver circuit **1021**, and the second scan signal line **1102** is electrically connected to the second driver circuit **1022**, thus ensuring the normal transmission of the scan signal and the display effect. The vertical projection of the first scan signal line **1101** on the plane where the substrate **101** is located and the vertical projection of the second scan signal line **1102** on the plane where the substrate **101** is located overlap with the vertical projection of the same second color-resist gap **1091** on the plane where the substrate **101** is located, that is, the first scan signal line **1101** and the second scan signal line **1102** overlap with the same second color-resist gap **1091** in the direction perpendicular to the plane where the substrate **101** is located. In this manner, the first scan signal line **1101** and the second scan signal line **1102** are directly connected to corresponding driver circuits, achieving a simple direct connection, reducing the difficulty of the preparation process and the preparation cost. FIGS. **20** to **27** only show one kind of connection between driver circuits and a scan signal line set as an example. The connection may be selected according to the actual design requirements and is not limited in the embodiments.
(90) In summary, the above-mentioned embodiments describe that a driver circuit set surrounds the signal lead-out structure. The adjustment of the position of driver circuits in the driver circuit set ensures that signal lead-out wires corresponding to multiple driver circuits are connected to corresponding first electrodes through the same insulating boss. On the premise of the normal transmission of drive signals is ensured, the coverage area of the color-resist layer is increased, and the display aperture ratio of the display panel can be increased in this arrangement.
(91) Next, the arrangement of a driver circuit set or signal lead-out wires are described in another embodiment in a case where at least two color-resist blocks surround the sidewall of the signal lead-out structure.
(92) As another feasible embodiment, FIG. **28** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIG. **28**, in some embodiments, the driver circuits **102** are arranged in an array.
(93) The signal lead-out wire **1042** includes a first lead portion **1045** and a second lead portion **1046** which are connected to each other.
(94) The first lead portion **1045** extends along a sidewall of the insulating boss **1041**, and the second lead portion **1046** is disposed in the same layer with the output electrode **103** and is electrically connected to the output electrode **103**.
(95) In a case where multiple driver circuits **102** share the same signal lead-out structure **104**, the signal lead-out wires **1042** are electrically connected to the output electrodes **103** of the driver circuits **102**. Since the driver circuits **102** may be arranged in a distribution manner according to the signal lead-out structure **104**, for a driver circuit **102** that is relatively far away from the signal lead-

out structure **104**, the signal lead-out wire **1042** may include the first lead portion **1045** and the second lead portion **1046** which are connected to each other. the first lead portion **1045** extends along the sidewall of the insulating boss **1041**, and the second lead portion **1046** and the output electrode **103** are disposed in the same layer and are electrically connected to each other, thus achieving the electrical connection between the output electrode **103** and the first electrode **106**. The second lead portion **1046** and the output electrode **103** are disposed in the same layer, thus ensuring the transmission of the drive signal and reducing the difficulty of the preparation process.
(96) Based on the above-mentioned embodiments, FIG. **29** is a structural diagram of an insulating boss according to an embodiment of the present disclosure. As shown in FIG. **29**, in some embodiments, the sidewall of the insulating boss **1041** includes multiple sub-sidewalls **120**: multiple sub-sidewalls **120** include at least a first sub-sidewall **1201**, a second sub-sidewall **1202**, and a third sub-sidewall **1203**, and the second sub-sidewall **1202** is connected to the first sub-sidewall **1201** and the third sub-sidewall **1203** separately.
(97) At least part of the first signal lead-out wire **1043** extends along the first sub-sidewall **1201**, and at least part of the second signal lead-out wire **1044** extends along the third sub-sidewall **1203**.
(98) The sidewall of the insulating boss **1041** may include multiple sub-sidewalls **120**, different signal lead-out wires **1042** extend along different sub-sidewalls, respectively, and at least one sub-sidewall **120** is disposed between different signal lead-out wires **1042**, thus avoiding crosstalk between signal lead-out wires **1042**, as shown in FIG. **29**. For example, in FIG. **29**, the sidewall of the insulating boss **1041** includes five sub-sidewalls **120**, the first signal lead-out wire **1043** extends along the first sub-sidewall **1201**, the second signal lead-out wire **1044** extends along the third sub-sidewall **1203**, and the second sub-sidewall **1202** is connected to the first sub-sidewall **1201** and the third sub-sidewall **1203** separately, thus ensuring the normal transmission of the signals of both the first signal lead-out wire **1043** and the second signal lead-out wire **1044**.
(99) Based on the above-mentioned embodiments, FIG. **30** is another structural diagram of an insulating boss according to an embodiment of the present disclosure. As shown in FIG. **30**, in some embodiments, the sidewall of the insulating boss **1041** includes multiple sub-sidewalls **120**, and the signal lead-out wire **1042** covers at least one sub-sidewall **120**.
(100) For example, as shown in FIG. **30**, the sidewall of the insulating boss **1041** includes five sub-sidewalls **120**, and a signal lead-out wire **1042** covers a sub-sidewall **120**. The signal lead-out wire **1042** covers the whole sub-sidewall **120**. Compared with the signal lead-out wire **1042** extending along a sub-sidewall **120** as a single line, the signal lead-out wire **1042** covering the whole sub-sidewall **120** can help to improve the production yield effectively, and the risk of wire breakage of the signal lead-out wire **1042** can be avoided. Meanwhile, the resistance of the signal lead-out wire **1042** is reduced, the loss of the signal on the signal lead-out wire **1042** during the transmission of the signal is reduced, and the transmission of the signal is improved.
(101) Based on the above-mentioned embodiments, FIG. **31** is another structural diagram of an insulating boss according to an embodiment of the present disclosure. As shown in FIG. **31**, in some embodiments, the insulating boss **1041** includes a first boss section end-surface **131** and a second boss section end-surface **132** which are disposed sequentially in the first direction (the direction X shown in the figure), and the first boss section end-surface **131** is located on a side of the second boss section end-surface **132** facing to the substrate **101**. A vertical projection of the first boss section end-surface **131** on the plane where the substrate **101** is located covers a vertical projection of the second boss section end-surface **132** on the plane where the substrate **101** is located.
(102) For example, as shown in FIG. **31**, taking the insulating boss **1041**, whose bottom is set as a pentagon and that includes five sub-sidewalls **120**, as an example, the insulating boss **1041** includes the first boss section end-surface **131** and the second boss section end-surface **132** which are disposed sequentially in the first direction, and the first boss section end-surface **131** is located on the side of the second boss section end-surface **132** facing to the substrate **101**. The vertical

projection of the first boss section end-surface **131** on the plane where the substrate **101** is located covers the vertical projection of the second boss section end-surface **132** on the plane where the substrate **101** is located. In this manner, the sub-sidewalls **120** of the insulating boss **1041** are ramp-shaped, so that the signal lead-out wires **1042** can be easy to lead out, thereby avoiding the risk of wire breakage of the signal lead-out wires **1042**.
(103) In some embodiments, FIG. **32** is another structural diagram of an insulating boss according to an embodiment of the present disclosure. As shown in FIGS. **31** and **32**, the insulating boss **1041** includes a coned boss structure as shown in FIG. **31**: or the sidewall of the insulating boss **1041** includes multiple sub-sidewalls, and at least one of the sub-sidewalls includes a curved surface as shown in FIG. **32**.
(104) As shown in FIG. **31**, the insulating boss **1041** includes a coned boss structure. The coned boss structure may include a circular coned boss structure with a circular bottom, a triangular coned boss structure with a triangular bottom, a square coned boss structure with a quadrilateral bottom, a pentagonal coned boss structure with a pentagonal bottom, and a hexagonal coned boss structure with a hexagonal bottom, and etc. The shape of the coned boss structure is not limited in the embodiments of the present disclosure. The pentagonal coned boss structure is only taken as an example in FIG. **31**. The sub-sidewall **120** of the insulating boss may include a curved surface. For example, as shown in FIG. **32**, the bottom of the insulating boss **1041** is hexagonal, the sidewall of the insulating boss **1041** includes six sub-sidewalls **120**, and each of the six sub-sidewalls **120** is a curved surface. When the signal lead-out wire **1042** extends along the sub-sidewall **120** that is a curved surface, the risk of wire breakage of the signal lead-out wires **1042** can be effectively reduced.
(105) FIG. **33** is another structural diagram of an array substrate according to an embodiment of the present disclosure. As shown in FIG. **33**, in some embodiments, the array substrate **100** further includes a planarization layer **140** which is disposed between a film where the driver circuits **102** are located and a film where the color-resist layer **105** is located, and the insulating boss **1041** and the planarization layer **140** are disposed in the same layer.
(106) The planarization layer **140** is located between the film where the driver circuits **102** are located and the film where the color-resist layer **105** is located, and the planarization layer **140** provides a planarized surface for the preparation of the color-resist layer **105**, thereby ensuring the preparation of the color-resist layer **105** with a planarized surface and ensuring the effect of the preparation of the display panel. The insulating boss **1041** and the planarization layer **140** are disposed in the same layer and prepared in the same process, so that the thickness of the array substrate can be effectively reduced, the preparation process of the array substrate can be simplified, and the efficiency of preparing the array substrate can be improved. The insulating boss **1041** and the planarization layer **140** are made of the same material, and different filling effects are used to show the structural differences between the insulating boss **1041** and the planarization layer **140** in the drawings.
(107) FIG. **34** is a top view of a color-resist layer according to an embodiment of the present disclosure. As shown in FIGS. **33** and **34**, in some embodiments, the array substrate **100** further includes a light-shielding layer **150** located on the side of the color-resist layer **105** facing away from the substrate **101**, and the light-shielding layer **150** covers the signal lead-out structure **104**.
(108) The light-shielding layer **150** is located on the side of the color-resist layer **105** facing away from the substrate **101**, and the color-resist layer **105** includes multiple color-resist blocks **1051**. The color-resist blocks **1051** may include red color-resist blocks **1052**, green color-resist blocks **1053**, and blue color-resist blocks **1054**. The distance between color-resist blocks **1051** of different colors is a thin line width **151**, and the distance between color-resist blocks **1051** of the same color is a thick line width **152**. The light-shielding layer **150** may be located at the thin line width **151** between color-resist blocks **1051** of different colors or may be located at the thick line width **152** between color-resist blocks **1051** of the same color, which is not limited in embodiments of the

present disclosure. In a case where the light-shielding layer **150** is located at the thick line width **152** between color-resist blocks **1051** of the same color, the light-shielding layer **150** covers the signal lead-out structure **104**. The light-shielding layer **150** is used to cover the gaps between the signal lead-out structure **104** and adjacent red color-resist blocks **1052**, the gaps between the signal lead-out structure **104** and adjacent green color-resist blocks **1053**, and the gaps between the signal lead-out structure **104** and adjacent blue color-resist blocks **1054**, thus reducing the light leakage of metal wires.
(109) In some embodiments, with further reference to FIGS. **2** and **33**, the first electrode **106** includes an electrode body **1061**, and the electrode body **1061** is electrically connected to the signal lead-out wire **1042**: or the first electrode includes an electrode body **1061** and an electrode bridging structure **1062**, and the electrode bridging structure **1062** is electrically connected to the signal lead-out wire **1042** and the electrode body **1061** separately.
(110) As shown in FIG. **2**, the first electrode **106** includes the electrode body **1061**, the electrode body **1061** may be a pixel electrode, and the first electrode **106** is electrically connected to the output electrode **103** through the signal lead-out wire **1042**. As shown in FIG. **33**, to avoid the problem of deep drilling between the first electrode **106** and a respective driver circuit **102**, the first electrode **106** includes the electrode body **1061** and the electrode bridging structure **1062**, and the electrode bridging structure **1062** is electrically connected to the signal lead-out wire **1042** and the electrode body **1061** separately, thus improving the stability of the connection between the electrode body **1061** and the signal lead-out wire **1042** and reducing the difficulty of the preparation process.
(111) Embodiments of the present disclosure further provide a method for preparing an array substrate, or that is to say, the array substrate provided in the preceding embodiments can be prepared using the method for preparing an array substrate. Thus, the method for preparing an array substrate also has the beneficial effects of the array substrate. Same reference can be made to the above explanation of the array substrate and are not repeated hereinafter.
(112) For example, FIG. **35** is a flowchart of a method for preparing an array substrate according to an embodiment of the present disclosure. Referring to FIG. **35**, the method for preparing an array substrate includes the following S**101**-S**105**.
(113) In S**101**, a substrate is provided.
(114) The substrate may be a rigid substrate or a flexible substrate, the material of the rigid substrate may be glass or a silicon wafer, and the material of the flexible substrate may be ultra-thin glass, metal foil, or polymeric plastic material.
(115) In S**102**, driver circuits are prepared on a side of the substrate, and the driver circuit includes an output electrode.
(116) In S**103**, a signal lead-out structure is prepared on a side of the driver circuits facing away from the substrate, and the signal lead-out structure includes an insulating boss and signal lead-out wires. The signal lead-out wire is electrically connected to the output electrode, and at least part of the signal lead-out wire extends along a sidewall of the insulating boss.
(117) In S**104**, a color-resist layer is prepared on the side of the driver circuits facing away from the substrate, and the color-resist layer surrounds a sidewall of the signal lead-out structure. In the first direction, the thickness of the color-resist layer is less than or equal to the extension length of the signal lead-out wire, and the first direction is perpendicular to a plane where the substrate is located.
(118) The signal lead-out structure is first prepared, and then the color-resist layer is arranged to surround the sidewall of the signal lead-out structure, so that the process of drilling via-holes in the color-resist layer is omitted, thus facilitating the amelioration of problems such as difficulty to drill via-holes on the color-resist layer, color-resist residues in the via-holes and the like, and improving the production yield.
(119) In S**105**, a first electrode is prepared on a side of the signal lead-out structure facing away

from the substrate, and the first electrode is electrically connected to the signal lead-out wire.
(120) The first electrode and the output electrode are electrically connected through the signal lead-out structure, thus ensuring that the drive signal outputted from the driver circuit is normally transmitted to the first electrode, thereby ensuring the display effect.
(121) According to the method for preparing an array substrate provided by the embodiment of the present disclosure, the signal lead-out structure is first prepared, and then the color-resist layer is prepared. In such preparation process, the signal lead-out structure is arranged to include the insulating boss and the signal lead-out wires, at least part of the signal lead-out wire extends along the sidewall of the insulating boss, and thus the electrical connection between the output electrode of the driver circuit and the first electrode is achieved, that is, the electrical connection between the output electrode and the first electrode is achieved without through a via-hole in the color-resist layer. The preparation method does not need to drill via-holes in the color-resist layer, and the preparation process is simple. The display is not affected by residues of color-resist in the via-holes, and the problem of wire breakage of metal is not caused by the excessive depth of the via-holes, thereby ensuring the normal display of the display panel.
(122) FIG. **36** is another flowchart of a method for preparing an array substrate according to an embodiment of the present disclosure. Referring to FIG. **36**, the method for preparing an array substrate includes the following S**201**-S**206**.
(123) In S**201**, a substrate is provided.
(124) In S**202**, driver circuits are prepared on a side of the substrate, and the driver circuit includes an output electrode.
(125) In S**203**, an insulating boss is prepared on a side of the driver circuits facing away from the substrate, and the vertical projection of the insulating boss on a plane where the substrate is located at least partially does not overlap with the vertical projection of the output electrode on the plane where the substrate is located.
(126) In S**204**, signal lead-out wires are prepared at least on a sidewall of the insulating boss.
(127) A signal lead-out structure includes the insulating boss and the signal lead-out wires, and the vertical projection of the insulating boss on the plane where the substrate is located at least partially does not overlap with the vertical projection of the output electrode on the plane where the substrate is located, that is, at least part of the output electrode is exposed from the insulating boss. In this manner, the electrical connection between the signal lead-out wire and the output electrode is easy to be implemented. At least part of the signal lead-out wire extends along the sidewall of the insulating boss, thus avoiding the risk of wire breakage of the signal lead-out wire.
(128) In S**205**, a color-resist layer is prepared on the side of the driver circuits facing away from the substrate, and the color-resist layer surrounds a sidewall of the signal lead-out structure. In the first direction, the thickness of the color-resist layer is less than or equal to the extension length of the signal lead-out wire, and the first direction is perpendicular to the plane where the substrate is located.
(129) In S**206**, a first electrode is prepared on a side of the signal lead-out structure facing away from the substrate, and the first electrode is electrically connected to the signal lead-out wire.
(130) According to the method for preparing an array substrate provided by the embodiments of the present disclosure, the vertical projection of the insulating boss on the plane where the substrate is located at least partially does not overlap with the vertical projection of the output electrode on the plane where the substrate is located, so that at least part of the output electrode are exposed from the insulating boss, thereby avoiding complex wires of the electrical connection between the signal lead-out wire and the output electrode and reducing the risk of wire breakage.
(131) FIG. **37** is another flowchart of a method for preparing an array substrate according to an embodiment of the present disclosure. Referring to FIG. **37**, in some embodiments, the color-resist layer includes multiple color-resist blocks; the driver circuits include multiple driver circuit sets, and each driver circuit set includes at least a first driver circuit and a second driver circuit; and the

first driver circuit includes a first output electrode, and the second driver circuit includes a second output electrode.
(132) The method for preparing an array substrate includes the following S**301**-S**305**.
(133) In S**301**, a substrate is provided.
(134) In S**302**, driver circuits are prepared on a side of the substrate, where the driver circuits include multiple driver circuit set, a driver circuit set at least includes a first driver circuit and a second driver circuit, the first driver circuit includes a first output electrode, and the second driver circuit includes a second output electrode.
(135) In S**303**, a signal lead-out structure is prepared on a side of the driver circuits facing away from the substrate and at a joint position of pre-preparation regions of at least two color-resist blocks; where the signal lead-out structure includes an insulating boss and signal lead-out wires, the signal lead-out wires include at least a first signal lead-out wire and a second signal lead-out wire, the first signal lead-out wire is insulated from the second signal lead-out wire, the first signal lead-out wire is electrically connected to the first output electrode, and the second signal lead-out wire is electrically connected to the second output electrode; and at least part of the first signal lead-out wire and at least part of the second signal lead-out wire each extends along a sidewall of the insulating boss.
(136) The signal lead-out wires include the first signal lead-out wire and the second signal lead-out wire which are insulated from each other, so that signal lead-out wires are electrically connected to driver circuits, respectively, thus achieving the normal transmission of drive signals.
(137) In S**304**, the color-resist blocks are prepared in pre-preparation regions of the color-resist blocks, where at least two color-resist blocks surround the sidewall of the signal lead-out structure, a color-resist layer including the color-resist blocks surrounds the sidewall of the signal lead-out structure, a thickness of the color-resist layer is less than or equal to an extension length of the signal lead-out wire in a first direction, and the first direction is perpendicular to a plane where the substrate is located.
(138) The color-resist blocks may include red color-resist blocks, green color-resist blocks, and blue color-resist blocks. The order of preparation may be selected according to the actual preparing requirements and is not limited in the embodiments of the present disclosure. The thickness of the color-resist layer is less than or equal to the extension length of the signal lead-out wires, thus avoiding via-hole drilling at the color-resist layer and improving the production yield.
(139) In S**305**, a first electrode is prepared on a side of the signal lead-out structure facing away from the substrate, and the first electrode is electrically connected to the signal lead-out wire.
(140) According to the method for preparing an array substrate provided by the embodiment of the present disclosure, at least two color-resist blocks surround the signal lead-out structure, and the color-resist layer surrounds the sidewall of the signal lead-out structure. In this manner, drilling via-holes are avoided, and the difficulty of the preparation is reduced. Meanwhile, one signal lead-out structure can correspond to multiple color-resist blocks, thereby reducing the number of the signal lead-out structures needs to be provided, and effectively improving the display aperture ratio of the display panel.
(141) Embodiments of the present disclosure further provide a display panel, and FIG. **38** is a structural diagram of a display panel according to an embodiment of the present disclosure. Referring to FIG. **38**, the display panel **200** includes an array substrate **100** and further includes an opposing substrate **300** disposed opposite to the array substrate **100**, and a display dielectric layer **400** is disposed between the array substrate **100** and the opposing substrate **300**. The display panel includes the array substrate provided in the above embodiments, and therefore, the display panel provided by the embodiment of the present disclosure also has the beneficial effects described in the above embodiments, which are not repeated herein.
(142) Embodiments of the present disclosure further provide a display device. FIG. **39** is a structural diagram of a display device according to an embodiment of the present disclosure. As

shown in FIG. **39**, the display device includes the display panel **200** in the above-mentioned embodiment. The display device includes the display panel described in any embodiment of the present disclosure. Therefore, the display device according to the embodiment of the present disclosure has corresponding effects of the display panel according to the embodiment of the present disclosure, which are not repeated herein. In an example, the display device may be a mobile phone, a computer, a smart wearable device (for example, a smart watch), an in-vehicle display device, and other electronic devices, which is not limited in the embodiments of the present disclosure.

Claims

1. An array substrate, comprising: a substrate; driver circuits located on a side of the substrate, wherein a driver circuit of the driver circuits comprises an output electrode; a signal lead-out structure, wherein the signal lead-out structure is located on a side of the driver circuits facing away from the substrate, the signal lead-out structure comprises an insulating boss and signal lead-out wires, a signal lead-out wire of the signal lead-out wires is electrically connected to the output electrode, and at least part of the signal lead-out wire extends along a sidewall of the insulating boss; a color-resist layer, wherein the color-resist layer is located on the side of the driver circuits facing away from the substrate, and the color-resist layer surrounds a sidewall of the signal lead-out structure, and wherein in a first direction, a thickness of the color-resist layer is less than or equal to an extension length of the signal lead-out wire, and the first direction is perpendicular to a plane where the substrate is located; and a first electrode, wherein the first electrode is located on a side of the signal lead-out structure facing away from the substrate, and the first electrode is electrically connected to the signal lead-out wire.
2. The array substrate according to claim 1, wherein the color-resist layer comprises a plurality of color-resist blocks, and at least one color-resist block of the plurality of color-resist blocks contacts the sidewall of the signal lead-out structure.
3. The array substrate according to claim 2, wherein at least two color-resist blocks of the plurality of color-resist blocks surround the sidewall of the signal lead-out structure; wherein the driver circuits comprise a plurality of driver circuit sets, a driver circuit set of the plurality of driver circuit sets comprises a first driver circuit and a second driver circuit, and a number of driver circuits in the driver circuit set is the same as a number of the at least two color-resist blocks surrounding the sidewall of the signal lead-out structure; wherein the first driver circuit comprises a first output electrode, and the second driver circuit comprises a second output electrode; wherein the signal lead-out wires comprise a first signal lead-out wire and a second signal lead-out wire, and the first signal lead-out wire is insulated from the second signal lead-out wire; and wherein the first signal lead-out wire is electrically connected to the first output electrode, and the second signal lead-out wire is electrically connected to the second output electrode.
4. The array substrate according to claim 3, wherein the plurality of driver circuit sets are disposed around the signal lead-out structure.
5. The array substrate according to claim 4, wherein the plurality of color-resist blocks are arranged in a matrix, the plurality of color-resist blocks comprise a plurality of color-resist columns, and a first color-resist gap exists between two adjacent color-resist columns of the plurality of color-resist columns; wherein the array substrate further comprises a plurality of data signal line sets, a data signal line set of the plurality of data signal line sets comprises a first data signal line and a second data signal line, the first data signal line is electrically connected to the first driver circuit, and the second data signal line is electrically connected to the second driver circuit; wherein a vertical projection of the first data signal line on the plane where the substrate is located and a vertical projection of the second data signal line on the plane where the substrate is located overlap vertical projections of different first color-resist gaps on the plane where the substrate is located,

respectively; and wherein the first data signal line further comprises a first data connection line, and the first driver circuit is electrically connected to the first data signal line via the first data connection line, or the second data signal line further comprises a first data connection line, and the second driver circuit is electrically connected to the second data signal line via the first data connection line.

6. The array substrate according to claim 4, wherein the plurality of color-resist blocks are arranged in a matrix, the plurality of color-resist blocks comprise a plurality of color-resist columns, and a first color-resist gap exists between two adjacent color-resist columns of the plurality of color-resist columns; wherein the array substrate further comprises a plurality of data signal line sets, a data signal line set of the plurality of data signal line sets comprises a first data signal line and a second data signal line, the first data signal line is electrically connected to the first driver circuit, and the second data signal line is electrically connected to the second driver circuit; and wherein a vertical projection of the first data signal line on the plane where the substrate is located and a vertical projection of the second data signal line on the plane where the substrate is located overlap with a vertical projection of a same first color-resist gap on the plane where the substrate is located.

7. The array substrate according to claim 6, wherein the first data signal line and the second data signal line are disposed in different layers; wherein the first data signal line is located on a side of the color-resist layer facing to the substrate, and the second data signal line is located on a side of the color-resist layer facing away from the substrate; wherein the second data signal line further comprises a second data connection line, the second data signal line is electrically connected to the second driver circuit via the second data connection line, and the second data connection line extends along a sidewall of the insulating boss; and wherein the insulating boss comprises a coned boss structure, or the sidewall of the insulating boss comprises a plurality of sub-sidewalls and at least one of the plurality of sub-sidewalls comprises a curved surface.

8. The array substrate according to claim 4, wherein the plurality of color-resist blocks are arranged in a matrix, the plurality of color-resist blocks comprise a plurality of color-resist rows, and a second color-resist gap exists between two adjacent color-resist rows of the plurality of color-resist rows; wherein the array substrate further comprises a plurality of scan signal line sets, a scan signal line set of the plurality of scan signal line sets comprises a first scan signal line and a second scan signal line, the first scan signal line is electrically connected to the first driver circuit, and the second scan signal line is electrically connected to the second driver circuit; wherein a vertical projection of the first scan signal line on the plane where the substrate is located and a vertical projection of the second scan signal line on the plane where the substrate is located overlap with vertical projections of different second color-resist gaps on the plane where the substrate is located, respectively; and wherein the first scan signal line further comprises a first scan connection line, and the first driver circuit is electrically connected to the first scan signal line via the first scan connection line; or the second scan signal line further comprises a first scan connection line, and the second driver circuit is electrically connected to the second scan signal line via the first scan connection line.

9. The array substrate according to claim 4, wherein the plurality of color-resist blocks are arranged in a matrix, the plurality of color-resist blocks comprise a plurality of color-resist rows, and a second color-resist gap exists between two adjacent color-resist rows; wherein the array substrate further comprises a plurality of scan signal line sets, a scan signal line set of the plurality of scan signal line sets comprises a first scan signal line and a second scan signal line, the first scan signal line is electrically connected to the first driver circuit, and the second scan signal line is electrically connected to the second driver circuit; and wherein a vertical projection of the first scan signal line on the plane where the substrate is located and a vertical projection of the second scan signal line on the plane where the substrate is located overlap with a vertical projection of a same second color-resist gap on the plane where the substrate is located.

10. The array substrate according to claim 3, wherein the driver circuits are arranged in an array;

wherein the signal lead-out wire comprises a first lead portion and a second lead portion which are connected; and wherein the first lead portion extends along a sidewall of the insulating boss, and the second lead portion is disposed in a same layer with the output electrode and is electrically connected to the output electrode.
11. The array substrate according to claim 3, wherein the sidewall of the insulating boss comprises a plurality of sub-sidewalls; wherein the plurality of sub-sidewalls comprise a first sub-sidewall, a second sub-sidewall, and a third sub-sidewall, and the second sub-sidewall is connected to the first sub-sidewall and the third sub-sidewall separately; and wherein at least part of the first signal lead-out wire extends along the first sub-sidewall, and at least part of the second signal lead-out wire extends along the third sub-sidewall.
12. The array substrate according to claim 1, wherein the sidewall of the insulating boss comprises a plurality of sub-sidewalls; and wherein the signal lead-out wire covers at least one of the plurality of sub-sidewalls.
13. The array substrate according to claim 1, wherein the insulating boss comprises a first boss section end-surface and a second boss section end-surface which are disposed sequentially in the first direction, and the first boss section end-surface is located on a side of the second boss section end-surface facing to the substrate; and wherein a vertical projection of the first boss section end-surface on the plane where the substrate is located covers a vertical projection of the second boss section end-surface on the plane where the substrate is located.
14. The array substrate according to claim 1, wherein the array substrate further comprises a planarization layer which is disposed between a film where the driver circuits are located and a film where the color-resist layer is located; and wherein the insulating boss and the planarization layer are disposed in a same layer; and wherein the array substrate further comprises a light-shielding layer which is located on a side of the color-resist layer facing away from the substrate; and wherein the light-shielding layer covers the signal lead-out structure.
15. The array substrate according to claim 1, wherein the first electrode comprises an electrode body, and the electrode body is electrically connected to the signal lead-out wire; or wherein the first electrode comprises an electrode body and an electrode bridging structure, and the electrode bridging structure is electrically connected to the signal lead-out wire and the electrode body separately.
16. A display panel, comprising the array substrate according to claim 1.
17. A display device, comprising the display panel according to claim 16.
18. A method for preparing an array substrate, comprising: providing a substrate; preparing driver circuits on a side of the substrate, wherein a driver circuit of the driver circuits comprises an output electrode; preparing a signal lead-out structure on a side of the driver circuits facing away from the substrate, wherein the signal lead-out structure comprises an insulating boss and signal lead-out wires, a signal lead-out wire of the signal lead-out wires is electrically connected to the output electrode, and at least part of the signal lead-out wire extends along a sidewall of the insulating boss; preparing a color-resist layer on the side of the driver circuits facing away from the substrate, wherein the color-resist layer surrounds a sidewall of the signal lead-out structure, and wherein in a first direction, a thickness of the color-resist layer is less than or equal to an extension length of the signal lead-out wire, and the first direction is perpendicular to a plane where the substrate is located; and preparing a first electrode on a side of the signal lead-out structure facing away from the substrate, wherein the first electrode is electrically connected to the signal lead-out wire.
19. The method according to claim 18, wherein the preparing the signal lead-out structure on the side of the driver circuits facing away from the substrate comprises: preparing the insulating boss on the side of the driver circuits facing away from the substrate, wherein a vertical projection of the insulating boss on a plane where the substrate is located at least partially does not overlap with a vertical projection of the output electrode on the plane where the substrate is located; and preparing the signal lead-out wire at least on the sidewall of the insulating boss.

20. The method according to claim 18, wherein the color-resist layer comprises a plurality of color-resist blocks; wherein the driver circuits comprise a plurality of driver circuit sets, and a driver circuit set of the plurality of driver circuit sets comprises a first driver circuit and a second driver circuit; wherein the first driver circuit comprises a first output electrode, and the second driver circuit comprises a second output electrode; and wherein the preparing the signal lead-out structure on the side of the driver circuits facing away from the substrate comprises: preparing the signal lead-out structure on the side of the driver circuits facing away from the substrate and at a joint position of pre-preparation regions of at least two color-resist blocks of the plurality of color-resist blocks, wherein the signal lead-out wires comprises a first signal lead-out wire and a second signal lead-out wire, the first signal lead-out wire is insulated from the second signal lead-out wire, the first signal lead-out wire is electrically connected to the first output electrode, and the second signal lead-out wire is electrically connected to the second output electrode; and wherein the preparing the color-resist layer on the side of the driver circuits facing away from the substrate comprises: preparing color-resist blocks in pre-preparation regions of the color-resist blocks of the plurality of color-resist blocks, wherein at least two color-resist blocks of the plurality of color-resist blocks surround the sidewall of the signal lead-out structure.